# Dell's Circular Economy Strategy for Sustainable Electronic Waste Management

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***Abstract*—Electronic waste is among the fastest-growing waste streams, and the linear take–make–dispose model of electronics manufacturing discards both devices and the materials embedded in them. This paper examines how Dell Technologies applies circular economy principles to electronic waste management and how far its publicly reported outcomes support its stated ambitions. Using a secondary-research case-study method, we combine documentary analysis of Dell's sustainability disclosures and policies with peer-reviewed literature, international e-waste statistics and Indian regulatory publications, and we compare the quantitative indicators reported across fiscal years. Six strategy areas are analysed: repairable and modular design, life extension and refurbishment, take-back and reverse logistics, recycling and closed-loop materials, packaging, and supply-chain collaboration. The evidence shows a coherent, hierarchy-based strategy and a steady rise in the share of recycled and renewable materials in products. However, reported product collection remains far below the company's 2030 take-back goal and fell in the latest year, and the disclosures say little about reuse volumes, repair outcomes or regional performance. Collection and recycling figures alone therefore cannot demonstrate circularity. We discuss practical challenges, draw implications for India's extended-producer-responsibility regime, and recommend consistently defined, independently assured and more transparent circularity reporting. The study uses no primary data, and its findings depend on the quality of corporate disclosures.**

***Keywords—circular economy, electronic waste management, Dell Technologies, sustainable product design, reverse logistics, resource recovery, recycling, extended producer responsibility***

## I. INTRODUCTION

Electrical and electronic equipment is now pervasive, and the volume discarded each year is growing faster than the systems built to manage it. The Global E-waste Monitor 2024 estimates that 62 million tonnes (Mt) of e-waste were generated worldwide in 2022, 82% more than in 2010, and that only 22.3% was formally documented as collected and recycled [1]. It projects generation of 82 Mt by 2030, with the documented collection-and-recycling rate falling to about 20% [1]. The same report values the metals embedded in 2022 e-waste at about US$91 billion, of which roughly US$62 billion was recoverable but unaccounted for in formal recovery [1].

The linear take–make–dispose lifecycle turns valuable materials into waste after a comparatively short period of use. Recycling recovers materials but not the manufacturing effort embedded in a working device, and its contribution to scarce-material supply is still small: the Monitor reports that e-waste recycling meets no more than 1% of rare-earth demand [1]. Improper disposal compounds the loss because informal dismantling, burning and landfilling forfeit the recoverable material while exposing workers and ecosystems to hazardous constituents.

The circular economy (CE) offers an alternative in which design, business models and recovery systems aim to keep products and materials in use at their highest value [2], [3]. For electronics this implies a hierarchy: durable and repairable design, repair and upgrade, reuse and refurbishment, component harvesting, and only then materials recycling [4], [5].

Dell Technologies is a suitable case for examining that hierarchy. It manufactures client and enterprise computing hardware, it has published circular economy principles with quantified 2030 goals [4], and it reports multi-year indicators against which progress can be checked [6]–[8]. Most available information on its circular outcomes is company-reported; in the sources we consulted we found few independent evaluations. This paper therefore treats the strategy as an analytical case rather than a set of commitments to be restated, and it addresses six research questions:

- *RQ1:* How are circular economy principles built into Dell's product design, manufacturing, use, recovery and end-of-life management?
- *RQ2:* How do take-back, repair, reuse, refurbishment and recycling partnerships contribute to e-waste management?
- *RQ3:* How does the use of recycled, renewable and recovered materials support resource conservation?
- *RQ4:* What measurable progress has Dell reported against its circular economy targets?
- *RQ5:* What limitations, challenges and implementation gaps affect the strategy's effectiveness?
- *RQ6:* What can other manufacturers and India's e-waste ecosystem learn from this approach?

The central argument is that Dell's strategy combines product design, life extension, take-back and materials recovery in a coherent way, but that its effectiveness must be judged against measured outcomes, implementation constraints, and the difference between stated commitments and demonstrated results. Section II reviews the literature, Section III explains the method, Section IV analyses the strategy, Section V evaluates reported performance, Sections VI and VII discuss challenges and implications for India, and Sections VIII and IX give recommendations and conclusions.

## II. Background and Related Literature

### A. Circular Economy Concepts

The circular economy has no single accepted definition. A review of 114 definitions found that they most often describe circularity as a combination of reduce, reuse and recycle activities, and that they differ widely in how far they address systemic change, business models and consumers [3]. The Ellen MacArthur Foundation popularised a framing in which the economy is restorative by design and products and materials are kept at their highest utility [2]. Geissdoerfer *et al.* argue that the relationship between circularity and sustainability is conditional rather than automatic [9]. Circular practices therefore need evidence of net outcomes, not only of activity, which is the lens applied in this study.

### B. Life Extension, Repair, Reuse and Refurbishment

Strategies that keep an existing device in service sit above recycling in circular hierarchies because they preserve the function of the device and not only its constituent materials. Dell's own documents rank them this way, from whole-system reuse through refurbishment and component reuse to commodity-material recovery [4], [5]. The academic literature supports this ordering: Reike *et al.* arrange resource-value-retention options from higher-value retention, such as reuse, repair, refurbishment and remanufacturing, down to recycling and recovery [10], and Bakker *et al.* discuss product life extension as a design-led strategy [11]. Effectiveness depends on repairability, meaning access to parts, documentation and non-destructive disassembly, and on what users do with devices they no longer need. Reviews of consumer behaviour indicate that disposal habits and awareness strongly influence how much equipment reaches formal channels [12], [13].

### C. Reverse Logistics and Closed-Loop Supply Chains

Reverse logistics covers the collection, transport, testing, sorting and processing of returned products. The literature treats these activities as part of closed-loop supply-chain design in which forward and return flows are planned together [14]. Returned devices differ in condition, so each must be assessed to decide between reuse, component recovery and recycling; the quality of that triage largely determines how much value is retained.

### D. Extended Producer Responsibility

Extended producer responsibility (EPR) holds the companies that place products on the market accountable for managing the waste those products eventually become [15]. It is the regulatory channel through which take-back is institutionalised in many jurisdictions, including India (Section VII).

### E. Contribution of This Study

Together, these sources establish the hierarchy of circular loops, the dependence of recovery on consumer behaviour and regulation, and the caution that circularity must be demonstrated by outcomes. In the sources reviewed, we found less on how a specific manufacturer's multi-year indicators should be read when definitions and scope change. This paper examines that question for Dell.

## III. Research Methodology

The study is a secondary-research, single-case design based on publicly available documents. No interviews, surveys, field visits, laboratory work or primary-data collection were undertaken, and no statistical tests were performed.

### A. Sources and Selection Criteria

Four source groups were used. (i) Company documents: the FY24 ESG report [6], the FY25 and FY26 "Impact by the Numbers" briefs [7], [8], the Circular Economy Principles [4], the Electronics Disposition Policy [5], the Producer Responsibility Policy [16] and a Concept Luna publication [17], all retrieved from Dell's public websites. (ii) Intergovernmental statistics: the Global E-waste Monitor [1]. (iii) Indian official publications: two Press Information Bureau releases of parliamentary replies [18], [19] and a Central Pollution Control Board (CPCB) guidance document [20]. (iv) Peer-reviewed and institutional literature [2], [3], [9]–[15], [21].

Sources were included if they (a) bore on at least one research question, (b) came from the company, an official body or a peer-reviewed outlet, (c) were the most recent disclosure for current status or an earlier one needed for trends, and (d) could be retrieved and read directly. Trade-media items about Dell's activities in India and references whose details could not be verified were excluded. The FY26 brief covers the fiscal year ended 30 January 2026 [8].

### B. Analytical Procedure

Qualitative thematic analysis proceeded by reading each document against a coding frame derived from the research questions: the six strategy areas in Section IV, the distinction between stated commitment and reported outcome, and the scope and assurance of each indicator. Quantitative indicators were extracted as reported, with units, fiscal year and source, and were compared across years only when the definition was the same. Derived quantities, such as percentage-point differences, were calculated by the authors from reported figures and are identified as derived.

### C. Limitations

Corporate disclosures are self-reported and selective, and assurance coverage varies. Indicator definitions can change between years, and the briefs give limited methodological detail. Dell's disclosures consulted here are global; regional results, including for India, were not available. Findings are limited to English-language, publicly accessible documents.

## IV. Dell Technologies' Circular Economy Strategy

Dell's principles adopt three framework aims: eliminate waste and pollution, circulate products and materials at their highest value, and regenerate nature [4]. Its focus areas are designing out waste and obsolescence, using renewable energy and materials, offering customers new ways to transition, and partnering broadly [4]. The 2030 goals are to reuse or recycle one metric ton of product for every ton purchased, to use recycled, renewable or reduced-carbon-energy material for more than half of product content, and to make all packaging from recycled or renewable material or to reuse it [4], [6]. In Dell's reporting, "2030" denotes the end of fiscal year 2031 [6]. Fig. 1

shows the generic loops used as the analytical frame, and Table I maps Dell's stated mechanisms to the evidence available. These are company-level policies; they do not imply that every product follows the same design, material or recovery route.

TABLE I. DELL'S CIRCULAR STRATEGY: MECHANISMS AND EVIDENCE

| Area | Dell-stated mechanisms | Evidence in consulted disclosures |
|---|---|---|
| Design and repairability | Designing out waste; modular designs; self-repair resources; Concept Luna [4], [6], [17] | Qualitative; no repair or lifetime metric |
| Reuse and refurbishment | Reuse-first hierarchy; Reconnect with Goodwill; Asset Recovery Services [5], [6] | Reused or resold share of collected weight (9–11%); no unit counts [7], [8] |
| Take-back | Free consumer return in many countries; 1 t per t goal [6], [16] | Quantified: 28.1–33.7% of shipped weight [7], [8] |
| Recycling and materials | Recycled, renewable and reduced-carbon content; recovered gold, carbon fibre [6] | Quantified: 14.1–19.3% content; avoided extraction not quantified |
| Packaging | Recycled, renewable or reused packaging [6] | 96.4% (FY24), 97.6% (FY26); reuse metric in development [8] |
| Collaboration and governance | Partner audits; chain of custody; waiver council; data sanitisation [5] | Policy-level; audit results not disclosed |

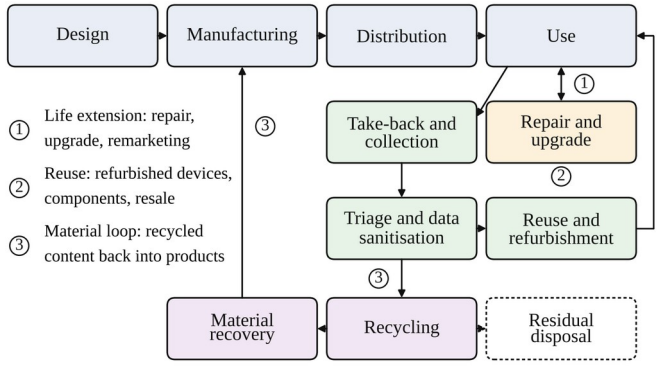



Fig. 1. Analytical schematic of circular loops across the electronics lifecycle, based on the hierarchy in [4], [5]. It is a generic frame, not a depiction of a specific Dell product or process.

### A. Circular Product Design and Repairability

Dell's principles define repairability and list designing out waste and obsolescence as a focus [4]. Its reporting cites modular designs and a self-repair resource site [6]. Concept Luna, introduced in fiscal year 2022 [6], is the most explicit design statement. Dell describes it as a forward-looking vision rather than a single shipped product, featuring modular construction with fewer adhesives and screws, usage telemetry, and recycled steel and aluminium [17]. The same source mentions more than 430 patent applications, not attributed solely to Luna, and gives no quantified savings or list of shipping products that embody its features [17].

The practical relevance is clear: easier disassembly can shorten repair and refurbishment times and improve material separation at end of life. However, the evidence is of design direction, not of deployed circularity, and none of the consulted briefs reports repair rates, parts availability or product lifetimes.

### B. Product Life Extension, Reuse and Refurbishment

The Electronics Disposition Policy sets a reuse hierarchy of whole-system reuse, refurbishment or remanufacture, component reuse and commodity-material recovery, with recycling last, and requires data sanitisation in line with NIST SP 800-88 and IEEE 2883 [5]. Dell's reporting also names Dell Reconnect with Goodwill, a partnership begun in 2004 in the United States, and Asset Recovery Services [6]. Quantitatively, the briefs report that 89.0% (FY25) and 91.0% (FY26) of collected product by weight was recycled, with the remainder reused or resold [7], [8]. They do not report the number of devices or components refurbished or resold, so the size of the reuse tier cannot be assessed beyond that residual share.

### C. Take-Back Programs and Reverse Logistics

The take-back goal is to reuse or recycle one metric ton for every ton of product purchased by 2030, measured by weight rather than unit count [6]. Dell reports cumulative recovery of more than 1.3 billion kg of used electronics since 2007 [6]. Annual collection was 27.3%, 30.1% and 33.7% of shipped weight in FY23–FY25 [7], and 28.1% in FY26, a figure that covers products received through all channels and not only Dell-owned ones [8]. Its Producer Responsibility Policy supports individual producer responsibility with internalised costs and offers free consumer take-back in many countries, while opposing government-collected fees for public infrastructure and visible consumer fees [16].

The disposition policy bars sending e-waste to landfills or incinerators and restricts exports from OECD and EU countries to non-OECD and non-EU countries unless national law allows them and Dell gives prior approval [5]. The indicators, however, are not broken down by region, customer type or channel, so the performance of the reverse-logistics network itself cannot be evaluated.

### D. Recycling, Material Recovery and Closed-Loop Materials

Recycled, renewable or reduced-carbon-energy material accounted for 14.1% of product content in FY24 [6], 17.4% in FY25 [7] and 19.3% in FY26 [8]. The total mass of such materials in products rose from about 39.2 million kg in FY23 to 58.6 million kg in FY26 [7], [8]. Dell cites closed-loop examples including recycled gold from motherboards (2018), reclaimed aerospace carbon fibre (from 2015), display steel with 50% recycled content and ocean-bound plastics [6]. Two cautions apply. First, the content goal was widened in FY23 to include reduced-carbon-energy materials [6], so the indicator is not a pure recycled-content measure. Second, a content share does not quantify avoided virgin extraction. Substituting secondary for virgin material conserves resources only to the extent that the secondary material would not otherwise have been used, and system-wide recovery remains modest for some critical materials [1].

### E. Sustainable Packaging and Resource Efficiency

The packaging goal covers recycled or renewable content and, since FY23, reused packaging, whose measurement was still in development [6]. Recycled or renewable content was 96.4% in FY24 [6] and 97.6% in FY26, the latter marked as third-party attested [8]; the FY25 brief gives no packaging figure [7]. Packaging therefore appears close to its ceiling, and the

unresolved element is reuse. For manufacturing waste, the FY26 brief reports 14,897 t, of which 6 t was hazardous [8], against 9,405 t in FY25 [7]. Neither brief explains the change in scope or cause, so we do not interpret it.

### *F. Stakeholder and Supply-Chain Collaboration*

Dell's principles call for partnering broadly [4]. In practice, the disposition policy relies on partner audits, chain-of-custody controls and a Return Stream Risk Governance Council that handles waivers [5]. The consulted version does not refer to third-party recycler certification schemes, and audit results are not disclosed, so the assurance behind these controls cannot be independently judged. Take-back partnerships such as Goodwill [6] extend reach, while the producer-responsibility policy shows engagement with regulators on how costs should be allocated [16].

## V. Performance Evaluation and Discussion

### *A. Reported Indicators and Progress*

Table II collates the circularity indicators that Dell reports for FY23–FY26, with targets and units. All values are company-reported. The FY24 figures appear in both the FY24 ESG report [6] and the FY25 brief [7], and they agree, which supports the internal consistency of the series. Four reporting differences limit comparison. First, take-back moved from a unit-count to a weight basis [6]. Second, goal definitions were widened in FY23 [6]. Third, the FY26 collection figure explicitly includes all channels [8], and the briefs do not state whether earlier years used the same scope. Fourth, assurance differs: limited assurance was commissioned for the FY23–FY25 metrics [7], whereas the FY26 collection rows carry no assurance marker, unlike the packaging figure [8]. The FY26 brief is also a single-year document without comparison columns, and it notes that Scope 3 methodology was refined without restating earlier years [8].

TABLE II. Reported Circularity Indicators, FY23–FY26

| Indicator (unit) | Target | FY23 | FY24 | FY25 | FY26 |
|---|---|---|---|---|---|
| Product collected (% of shipped weight) | 100% by 2030 | 27.3 | 30.1 | 33.7 | 28.1 |
| Collected product recycled (% of collected weight) | None stated | — | — | 89.0 | 91.0 |
| Recycled, renewable or reduced-carbon content (% of product) | >50% by 2030 | — | 14.1 | 17.4 | 19.3 |
| Sustainable materials in products (million kg) | None stated | 39.2 | 43.1 | 52.0 | 58.6 |
| Packaging recycled or renewable content (%) | 100% by 2030 | — | 96.4 | — | 97.6 |

Company-reported. "—": not reported in the documents consulted. Sources: FY23–FY25 [7]; FY24 values and targets [6]; FY26 [8]. Packaging target includes reused packaging. Scope and assurance differ between years (Section V-A).

Within these limits, progress is uneven. Product content rose by 5.2 percentage points (pp) between FY24 and FY26 (derived), about 2.6 pp per year, against a goal of more than 50% by 2030. Reaching it by the end of FY31 would require about 6.1 pp per year from the FY26 level (derived arithmetic, not a forecast). Collection rose to 33.7% in FY25 but fell by 5.6 pp to 28.1% in FY26, leaving a gap of 71.9 pp to the goal. The briefs do not explain the fall. Because the denominator is the weight of units shipped, growth in shipments or a shift toward heavier products could depress the ratio even if the weight collected were unchanged, but this is a hypothesis that the disclosures do not test. The recycled share of collected product rose by 2.0 pp to 91.0%, and packaging content rose by 1.2 pp between FY24 and FY26.

### *B. Collection, Reuse and Recycling Are Different Outcomes*

The take-back goal counts product that is "reused or recycled" without weighting the two routes, yet they sit at different levels of the hierarchy. If the FY26 percentages refer to the same base, about 25.6% of shipped weight (28.1% × 91.0%) was recycled and about 2.5% (28.1% × 9.0%) was reused or resold (derived). On that reading the highest-value route is a very small fraction of shipped weight. The briefs also do not state whether "recycled" means mass delivered to recyclers or mass that became recovered material, and yields by material are not reported. Collection indicates that a device entered the recovery system, recycling indicates processing, and only reuse and verified material recovery indicate that value was retained. The disclosures can be read as a ladder of evidence: stated commitments, activities, outputs such as tonnes collected, and outcomes such as avoided virgin extraction or longer service lives. The consulted documents cover the first three rungs but say little about the fourth.

### *C. Environmental Benefits and Operational Considerations*

Circular practices can reduce demand for virgin materials and extend the use of embedded manufacturing effort, but net benefits depend on whether recovered products and materials displace new production [9]. Dell reports Scope 3 Category 12 (end-of-life treatment of sold products) emissions of 289,400 $tCO_2e$ in FY26, 59.4% above its FY20 base year [8]. The brief does not attribute this to volume growth, methodology or end-of-life routes, so it should not be read as a circularity result in either direction. Operationally, reuse requires testing, data sanitisation and resale channels, while recycling depends on partner capacity. The global context shows the value at stake: about US$62 billion of metals in 2022 e-waste was recoverable but unaccounted for [1].

### *D. Strengths and Shortcomings*

The strategy's strengths are a hierarchy-based framework, quantified goals with a target year, multi-year disclosure with consistent overlapping figures, and policies on exports, data sanitisation and partner oversight [4]–[7]. Its shortcomings in the evidence are that collection is far from the goal and fell in the latest year, no interim milestones appear in the consulted briefs, reuse and repair outcomes are not reported, assurance coverage is uneven, recycling yields are undefined, and independent evaluation is scarce. These shortcomings concern disclosure and demonstrated outcomes, and they do not show that the underlying practices are ineffective.

## VI. Challenges and Opportunities

### *A. Challenges*

- *Collection and participation.* Return rates depend on household and enterprise behaviour as well as on programme design [12], [13]. Globally, only 22.3% of e-

waste was formally documented as collected and recycled [1].

- *Durability, repair and refurbishment economics.* Repair and refurbishment compete with new-device pricing. The consulted sources give no cost or lifetime data, so this is an analytical point, not an evidence-based finding.
- *Material separation and infrastructure.* Recovery depends on dismantling and processing capacity; recycling meets no more than 1% of rare-earth demand [1].
- *Supply-chain coordination.* Multi-tier chains of custody must be audited and exports controlled [5].
- *Data security.* Returned devices can hold personal and corporate data, which is why sanitisation to recognised standards precedes reuse [5].
- *Economics of recovered materials.* Value remains unrealised where recovery is informal or incomplete [1].

### B. Opportunities

The following are analytical opportunities and recommendations, not descriptions of existing Dell practice. Better traceability of devices and returns would let firms report reuse and recovery on a verifiable basis. Wider repair access, through parts, documentation and service networks, would extend lifetimes. Expanded refurbishment capacity and consolidated return logistics would raise the share of devices reused. Reporting recovery efficiency by material would show whether recycled mass becomes usable material. Transparent reporting with stable definitions and restated prior years would make progress verifiable.

## VII. Implications for India's E-Waste Management Ecosystem

### A. Regulatory Context

The E-Waste (Management) Rules, 2022 were notified in November 2022, came into force on 1 April 2023 and replaced the 2016 rules [18]. They require registration on the CPCB portal, provide for environmental compensation and audits, and allow collection and processing only by registered entities [18]. A centralised EPR portal, quarterly and annual returns, and State Pollution Control Board drives against informal processing support enforcement [19]. According to CPCB guidance, EPR recycling targets are 60% for FY2023–24 and 2024–25, 70% for 2025–26 and 2026–27, and 80% from 2027–28, expressed as shares of waste generation; recyclers generate EPR certificates for gold, copper, aluminium and iron, and refurbishers must register and issue a refurbishing certificate [20].

### B. Scale and Informal-Sector Context

CPCB estimates cited in Parliament put generation at about 1.01 Mt in FY2019–20 (22.07% collected or recycled), 1.35 Mt in FY2020–21 (26.31%) and 1.60 Mt in FY2021–22 (32.92%) [18]. Later estimates are 1.25 Mt for FY2023–24 and 1.40 Mt for FY2024–25 [19]. The drop in estimated generation between FY2021–22 and FY2023–24 is not explained in the sources consulted and may reflect changes in estimation method, so we do not read it as a real decrease. These collection shares are also not directly comparable with the global 22.3% figure because definitions differ. India-focused reviews discuss the persistence of informal handling alongside the formal system and the importance of consumer awareness [12], [21].

### C. Transferable Practices and Required Adaptation

*Transferable:* (i) A reuse-before-recycling hierarchy fits India's registration of refurbishers, but EPR certificates are tied to recovered metals [20], which rewards recycling output over reuse. A complementary indicator for refurbished units would be needed (our analysis). (ii) Dell's policy of individual producer responsibility with internalised costs [16] is consistent in orientation with India's producer-obligation model [19]; operational alignment was not assessed. (iii) Standards-based data sanitisation [5] could build trust in formal return channels. (iv) Weight-based, multi-year reporting with stated scope would make EPR performance comparable across producers.

*Requiring adaptation:* Free consumer take-back in many countries [16] presupposes accessible collection points and participation, which the Indian literature identifies as constraints [12], [21]. Reaching households and small enterprises, integrating informal collectors into registered chains, and ensuring price signals that favour formal recovery would need India-specific design. Dell's consulted disclosures do not present India-specific programmes or results, so this paper makes no claim about Dell's performance in India; the discussion concerns transferability only.

## VIII. Recommendations

*R1. Separate and define the three outcomes.* Manufacturers should report collection, reuse or resale, and recycling separately, with stated scope, units and denominators, and should restate prior years when definitions change. This addresses the comparability limits in Section V.

*R2. Extend independent assurance.* All circularity indicators, including collection and recycled share, should receive the same assurance as other headline metrics, because uneven coverage weakens verification.

*R3. Report reuse and repair.* Disclose devices and components refurbished or resold, repair rates and parts availability, so that the top of the hierarchy can be assessed.

*R4. Set interim milestones and disaggregate.* Publish interim targets towards 2030 and break collection down by region, customer type and product category to explain year-to-year movements such as the FY26 decline.

*R5. Disclose recovery yields.* Report recovered material by type, not only mass sent to recyclers, and the audit results behind partner oversight.

*R6. Align Indian EPR with the hierarchy.* Policymakers could pair recycler certificates with verifiable refurbishment records, standards-based data sanitisation and consistent producer reporting, while extending convenient collection to households and small enterprises.

*R7. Evaluate outcomes independently.* Researchers should test claimed benefits with life-cycle and field studies, since circularity does not automatically deliver sustainability [9].

## IX. Conclusion

This paper examined Dell's circular economy strategy through secondary evidence. Regarding RQ1 and RQ2, Dell's documents embed a reuse-first hierarchy and combine design intent, take-back, refurbishment channels and controlled recycling, although design evidence is largely conceptual and reuse is unquantified. For RQ3, recycled, renewable and reduced-carbon content grew from 14.1% to 19.3% of product content in FY24–FY26, but resource savings are not quantified. For RQ4, progress is mixed: content and packaging indicators advanced, while collection of 28.1% in FY26 is far from the 100% goal and below the FY25 level. For RQ5, the main gaps lie in disclosure, namely changing scope, uneven assurance, undefined recycling yields and absent reuse and repair data. For RQ6, the lessons are to define and separate outcomes, assure them independently, and design policy so that reuse is rewarded alongside recycling; the Indian regime already provides registration and certificate mechanisms on which such refinements could build.

The significance of the case is that circularity claims should be evaluated at the level of outcomes, not commitments. The study is limited by its reliance on corporate and official documents, the absence of primary data and of regional results for India, and the lack of independent assessments of Dell's results. Future work could combine product-level life-cycle data, return-stream audits and Indian field studies to test the benefits that corporate indicators imply.

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